

Periods and L -functions of meromorphic modular forms via finite contours

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Abstract

In this note, we assign to every meromorphic modular form f of weight k on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ a finite L-integral, which we denote $f \mapsto L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$; here τ_0 is a base-point in the upper half-plane, and $\gamma_S(\tau_0) : -\frac{1}{\tau_0} \rightarrow \tau_0$ and $\gamma_T(\tau_0) : \tau_0 - 1 \rightarrow \tau_0$ denote integration contours. These L-integrals are finite for all base-points and contours. For the special case of weakly holomorphic modular forms, i.e. where f is either bounded at the cusp or has poles exclusively at the cusp, this L-integral is uniquely defined, i.e. it is fully independent of τ_0 and is fully independent of $\gamma_S(\tau_0)$ and $\gamma_T(\tau_0)$. We prove that this finite L-integral, as we define it, exactly matches the L-integral of holomorphic modular forms, and, separately, that it equals the non-holomorphically regularized L-integrals of [1]. When f has poles in away from the cusp, i.e. inside the fundamental domain, $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ is unique and constant for paths within the same homotopy class; L^* 's depending on paths outside the same homotopy class differ by well-defined wall-crossing residues. We prove that they match the results in [2] for L-functions for meromorphic modular forms. Finally, we show a slight modification of this linear function of f yields its associated period polynomial when k is such that $\dim(S_k) = 1$. Our period polynomials exactly reproduce the numerical periods reported in [3]. All results are both proven analytically and supported, where relevant, by our associated python script [4].

1 Introduction and main results

In this paper, we introduce a general framework that defines L-functions for generic modular forms on the full group $\mathrm{SL}_2(\mathbb{Z})$, and extracts their periods for the explicit case when the holomorphic cusp forms are one-dimensional, i.e. when $\dim(S_k) = 1$. Classically, L-functions and periods are both well-defined and well-studied for holomorphic modular forms. However, there is a growing body of work exploring L-functions [1] and periods [3] for weakly holomorphic modular forms. Similarly, there are natural physical motivations to consider L-functions for properly meromorphic modular forms, which have poles *within* the fundamental domain [2].

In these works, and specifically in [1, 2], and throughout physics, divergences due to singularities in quantities of interest necessitate the introduction of regulators in order to define finite quantities. Regulators, though, often break symmetries, and care is needed to check that the regulator has not injected undue violence to the system. Of relevance here is that [1] introduced a non-holomorphic regulator to deal with the singularities at either end of the classical Mellin contour, $\int_0^\infty$, sufficient to regulate divergences at the cusps, but which violates holomorphicity in the deformed contour-integral, ruining path-dependence. Simultaneously, [2] concerned itself with L-functions for modular forms which diverge within the interior of the fundamental domain. Here, holomorphic dependence on τ is paramount, which puts this treatment at odds with the regulator in [1].

In this short note, we introduce manifestly finite contour integral functionals to define L-functions for generic meromorphic modular forms that are of sufficient generality to handle any

combination of poles at cusps and poles within the fundamental domain. Our construction is motivated by, and makes precise contact with, work which extracts period-coefficients from weak cusp forms [3]. Our integrals are manifestly finite throughout, and reproduce all results in the literature, unifying the quasi-period results and the L-function extensions within one concrete, natural framework.

What we construct. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, let f be a meromorphic modular form of even weight $k \in \mathbb{Z}$ on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint with the property that the pair of line segments

$$S\text{-segment: } [-1/\tau_0, \tau_0], \quad T\text{-segment: } [\tau_0 - 1, \tau_0] \quad (1)$$

in $\mathbb{H}$ avoids the Γ -orbit of the poles of f in $\mathbb{H}$. Define

$$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) := i^{-s} \left[\int_{\gamma_S(\tau_0)} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_T(\tau_0)} f(\tau) \tilde{k}_T(\tau, s) d\tau \right], \quad (2)$$

where the integration contour $\gamma_S(\tau_0)$ runs between $\gamma_S(\tau_0, 0) = -1/\tau_0$ and $\gamma_S(\tau_0, 1) = \tau_0$, the integration contour $\gamma_T(\tau_0)$ runs between $\gamma_T(\tau_0, 0) = \tau_0 - 1$ and $\gamma_T(\tau_0, 1) = \tau_0$ and the T -kernel

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1) \quad (3)$$

is built from two Hurwitz-zeta functions. When the contour avoids poles in f , both contour integrals contributing to $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ in (3) are finite for every basepoint $\tau_0 \in \mathbb{H}$ admissible for f and every $s \in \mathbb{C}$; no regulator, asymptotic-subtraction argument, or analytic-continuation step is required. Further, as we will show, $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ is invariant under both τ_0 -variations and $\gamma_S(\tau_0), \gamma_T(\tau_0)$ -variations, as long as neither $\gamma_S(\tau_0)$ nor $\gamma_T(\tau_0)$ cross any pole location within the fundamental domain.

We will show that $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ is the classical L-function for $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Further, because of its τ_0 -independence and finiteness for $f \in M_k^!(\mathrm{SL}_2(\mathbb{Z}))$, and because we reproduce the analytic formulae in [1], we unambiguously show that their candidate L-function is the canonical L-function one would associate with any weak modular form, $f \in M_k^!$.

Three theorems. The functional $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ obeys three properties that together identify it as the canonical finite-contour L-function of f :

- (i) (*Theorem 1.1*) When $\dim S_k = 1$, the integer- s specialisations of (2) reduce to a polynomial-kernel form that produces the periods $\omega^\pm(f)$ and quasi-periods $\eta^\pm(f)$ as an explicit linear functional in the Fourier coefficients of f . For $f = \Delta$ at $k = 12$ this recovers the Brown values of $\omega^\pm(\Delta)$; for $f = \hat{\Delta}$ this gives the quasi-periods $\eta^\pm(\hat{\Delta})$ from a finite-endpoint contour pair, with no divergence at the cusp.
- (ii) (*Theorem 1.2*) For $f \in M_k^!$, $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ is τ_0 -independent on $\mathbb{H}$ and produces the completed L-function $L^*(f, s)$ at every $s \in \mathbb{C}$ as a single closed-form incomplete-gamma sum, agreeing term by term in the Fourier expansion with [1]. Our proof in fact exploits the simplicity at the elliptic fixed-point basepoint $\tau_0 = i$, where the S -segment collapses to a point and (2) reduces to a single integral on the compact contour $[i-1, i]$.
- (iii) (*Theorem 1.3*) For general meromorphic f , $L^*(f, s; \gamma_S, \gamma_T)$ depends on (γ_S, γ_T) only through their homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$, with explicit wall-crossing residue jumps when the homotopy class changes by crossing a pole. Restricted to forms regular at the cusp, this reproduces the deformed-Mellin L-function $L_f^*(s)_{\mathrm{int}}$ (Corollary 1.4; see [2]).

The complete statements are below; their proofs occupy §3–§5.

Theorem 1.1 (Periods and quasi-periods via a linear functional, $\dim S_k \leq 1$). *Let $k \in 2\mathbb{Z}$ with $\dim S_k \leq 1$, and let $f \in S_k^!$. For the substantive cases $k \in \{12, 16, 18, 20, 22, 26\}$ where $\dim S_k = 1$, there exist universal polynomial kernels $k_S(j, \tau), k_T(j, \tau) \in \mathbb{Q}[\tau]$ for $j \in \{2, 3, \dots, k-2\}$, with k_S the bare monomial*

$$k_S(j, \tau) = \tau^{j-1} \quad (4)$$

and $k_T(j, \tau)$ a polynomial of degree at most $k-2$ built from the Bernoulli polynomials $B_m(\tau+1)$, $m = 1, 2, \dots, k$, such that

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1}}{\binom{k-2}{j}} \left[\int_{\gamma_S(\tau_0)} f(\tau) k_S(j, \tau) d\tau + \int_{\gamma_T(\tau_0)} f(\tau) k_T(j, \tau) d\tau \right] \quad (5)$$

for every $f \in S_k^!$, every $\tau_0 \in \mathbb{H}$, and every $j \in \{2, \dots, k-2\}$. The right side of (5) is basepoint- and monomial-index-independent (lockstep). The kernels $k_S(j, \tau)$ and $k_T(j, \tau)$ are the integer- $s = j$ specialisations of τ^{s-1} and $\tilde{k}_T(\tau, s)$ respectively, via $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$).

Higher-dimensional S_k . For $k \geq 24$ with $\dim S_k \geq 2$, the universal polynomial kernels $k_S(j, \tau), k_T(j, \tau)$ become matrix-valued: each $j \in \{2, \dots, k-2\}$ gives a $\dim S_k$ -component linear functional on $\{c_f(n)\}$ rather than a scalar. The structural argument behind Theorem 1.1 — coboundary inversion on V_{k-2} via δ_T^{-1} in closed Bernoulli-operator form — generalises without conceptual change, but the explicit kernel tables grow with $\dim S_k$, and the lockstep / basepoint-independence statement becomes a matrix identity. Working out the explicit matrix-valued kernels for $k = 24, 28, 30, \dots$ and connecting them to the existing Eichler–Shimura matrix-period basis literature is the natural generalisation of Theorem 1.1, and we leave it to a follow-up. Theorems 1.2 and 1.3 are unaffected and apply at every even k .

Theorem 1.2 (L -function for weakly holomorphic forms). *Let $k \in 2\mathbb{Z}$ and $f \in M_k^!$. With $\tilde{k}_T(\tau, s)$ the Hurwitz-zeta kernel of (3), for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,*

$$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) := i^{-s} \left[\int_{\gamma_S(\tau_0)} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_T(\tau_0)} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (6)$$

is finite and independent of both $\tau_0 \in \mathbb{H}$ and of the admissible contour choices $\gamma_S(\tau_0), \gamma_T(\tau_0)$. We write $L^(f, s)$ for the common value.*

Specialising to the elliptic fixed point $\tau_0 = i$ of the modular involution $S: \tau \mapsto -1/\tau$, the S -segment $[-1/i, i]$ collapses to the single point i , the S -integral in (6) vanishes, and L^ reduces to a single integral on the compact contour $[i-1, i] \subset \mathbb{H}$, with closed-form sum*

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (7)$$

The right side of (7) is the completed L -function $L^(f, s)$ of f : on holomorphic cusp forms it reduces to the classical Hecke–Weil Mellin transform, and on $M_k^!$ its equality to the formula of [1] proves that the incomplete-gamma construction — previously obtained via non-holomorphic regulator deformations — is the unique canonical L -function for every weakly holomorphic modular form.*

Classical Mellin recovery and canonical uniqueness. For $f \in S_k$ a holomorphic cusp form, taking the cusp-limit $\tau_0 \rightarrow i\infty$ in (6) collapses the S -segment onto the classical Mellin contour $[0, i\infty]$ while the T -segment vanishes exponentially (since $f(\tau) = O(e^{-2\pi \text{Im } \tau})$ on the segment as $\text{Im } \tau \rightarrow \infty$):

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) = \int_0^\infty t^{s-1} f(it) dt, \quad (8)$$

the classical completed L -integral of f . For $f \in M_k$ with $c_f(0) \neq 0$ the same limit holds after a standard constant-term subtraction (see [2, Lemma 2.1] for the explicit polynomial correction).

For $f \in M_k^!$ with cusp poles the cusp-limit is genuinely pathological: $f(\tau) \rightarrow \infty$ as $\text{Im } \tau \rightarrow \infty$, so the S -segment integral diverges along the imaginary axis, and the T -segment integral $\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau$ acquires a matching divergence of the opposite sign as $\text{Im } \tau_0 \rightarrow \infty$. The two divergences cancel at every finite τ_0 , so $L^*(f, s)$ is finite and well defined throughout $\mathbb{H}$, but no naive Mellin contour limit exists. The construction of [1] recovers $L^*(f, s)$ on $M_k^!$ by a non-holomorphic regulator at the cusps; Theorem 1.2 shows that this regulator and the finite-contour functional (6) compute the same object, term by term in the Fourier expansion. In particular, $L^*(f, s)$ is independent of any choice of regularisation: it is the value of the finite-contour functional at any admissible $(\gamma_S(\tau_0), \gamma_T(\tau_0))$, and the BFK construction is one closed-form evaluation of it (at $\tau_0 = i$).

Theorem 1.3 (*L -function for meromorphic modular forms*). *Let $k \in 2\mathbb{Z}$ and let f be a meromorphic modular form of weight k on Γ . Write $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ for the upper half-plane with the Γ -orbit of the interior poles of f removed, and call the pair $(\gamma_S(\tau_0), \gamma_T(\tau_0))$ admissible if $\tau_0 \in \mathbb{H}_f$ and both contours lie in $\mathbb{H}_f$, with $\gamma_S(\tau_0)$ running from $S^{-1}\tau_0 = -1/\tau_0$ to τ_0 and $\gamma_T(\tau_0)$ from $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 . Define $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ by (2) with the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3). Then:*

- (a) *Both integrals in (2) are finite for every admissible $(\gamma_S(\tau_0), \gamma_T(\tau_0))$ and every $s \in \mathbb{C}$.*
- (b) *$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ depends on the admissible pair only through its homotopy class in $\mathbb{H}_f$: any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint τ_0 — leaves the value unchanged.*
- (c) *If two admissible configurations $(\gamma_S(\tau_0), \gamma_T(\tau_0))$ and $(\gamma_S(\tau'_0), \gamma_T(\tau'_0))$ lie in homotopy classes related by a single crossing of an interior pole $\tau_p \in \mathbb{H}$ of f , then*

$$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) - L^*(f, s; \gamma_S(\tau'_0), \gamma_T(\tau'_0)) = \mathcal{R}_{\tau_p}(f, s), \quad (9)$$

where for a simple pole $\text{Res}(f, \tau_p)$,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \text{Res}(f, \tau_p) \left[\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (10)$$

with $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ the winding-number contributions of γ_S and γ_T across τ_p (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in τ_p of (10); the closed form is given in §5.

Corollary 1.4 (*L^* on regular-at-cusps meromorphic forms*). *Let $f \in F_k$ be a meromorphic modular form of weight k that is regular at the cusp. Let $L_f^*(s)_{\text{int}, \gamma}$ denote the deformed-Mellin L -function of f along a contour γ from 0 to $i\infty$ lying in $\mathbb{H}_f$ (see [2, §2]). Then for every admissible $(\gamma_S(\tau_0), \gamma_T(\tau_0))$ whose homotopy class in $\mathbb{H}_f$ is compatible with γ (in the sense that the contour $\gamma_S(\tau_0)$ together with the cusp-limit piece $\lim_{\tau_0 \rightarrow i\infty} \gamma_S(\tau_0) \rightarrow [0, i\infty]$ can be deformed to γ in $\mathbb{H}_f$),*

$$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) = L_f^*(s)_{\text{int}, \gamma} \quad \text{for every } s \in \mathbb{C}. \quad (11)$$

The structural picture. The contour pair (1) has two natural one-parameter specialisations of the basepoint, each exploiting a fixed point of the modular curve $Y(1)$: the cusp specialisation $\tau_0 \rightarrow i\infty$ sends the S -segment to the classical Mellin contour $[0, i\infty]$ and the T -segment to 0 exponentially; the elliptic specialisation $\tau_0 = i$ kills the S -segment identically and leaves L^* as a single integral on $[i-1, i]$. The cusp specialisation, applied at finite $\tau_0 = i\Lambda$, gives the Λ -truncated Mellin contour of $L_f^*(s)_{\text{int}}$ (see [2]) as the S -segment alone, with the T -segment serving as the missing ingredient that produces the cocycle and renders the $\Lambda \rightarrow \infty$ limit unnecessary. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ in (3) is the closed form of the T -segment kernel: it is the unique V_n -analog, valid at every $s \in \mathbb{C}$, of the polynomial coboundary inversion of [7] (which itself recovers as the integer- s specialisation, by Theorem 1.1).

Outline. §2 introduces the period-defining cocycle and its moment expansion in the polynomial coefficient system V_{k-2} . §3 derives the polynomial kernels k_S^j, k_T^j and proves Theorem 1.1. §4 derives the Hurwitz-zeta kernel $\tilde{k}_T$ via a coboundary equation, proves Theorem 1.2 via a Hurwitz-unfolding lemma, and contains an explicit worked example at $k = 12$ exhibiting the Bernoulli-polynomial collapse of $\tilde{k}_T$ at integer s together with numerical values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$. §5 proves Theorem 1.3 and Corollary 1.4. §6 collects conceptual framings (holonomy/Wilson-loop interpretation, on-shell-amplitudes analogy, the regularisation-as-symptom heuristic, the open question on the geometric content of the $\tau^{s-1} f d\tau$ Mellin kernel, and pointers to the sequel on systematic per-mode theory for interior-pole forms). Appendix A records the algebraic derivation of the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ from contour-invariance of the two-segment functional: starting from the classical Mellin transform split at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding that the resulting incomplete-gamma tail be representable as a T -segment integral, one derives a shift-difference equation whose solution is the Hurwitz-zeta combination of (3). Appendix B is a brief Eichler–Shimura primer for readers approaching the subject from theoretical physics. This appendix is included for the working-version of this note and is intended to be removed before formal publication.

2 The period-defining cocycle and its moment expansion

The polynomial coefficient system. Let $V_n := \mathbb{C}[X, Y]_n$ be the space of complex homogeneous polynomials in two auxiliary variables X, Y of degree n , equipped with the right Γ -action

$$(P|_\gamma)(X, Y) = P(aX + bY, cX + dY) \quad \text{for } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma.$$

For a meromorphic modular form f of weight k , write $n := k-2$ throughout and form the V_n -valued 1-form on $\mathbb{H}$

$$\omega_f(\tau) := f(\tau) (X - \tau Y)^n d\tau.$$

Modularity of f implies that ω_f is Γ -equivariant in the standard sense [12]: under $\tau \mapsto \gamma^{-1}\tau$, the weight- k transformation of f cancels the weight- $(-n)$ slash on $(X - \tau Y)^n$, so the V_n -valued 1-form ω_f is invariant under the simultaneous action of Γ on τ and on the auxiliary variables (X, Y) . Equivalently, ω_f is a well-defined V_n -valued 1-form on the modular curve $Y(1) = \Gamma \backslash \mathbb{H}$ (modulo the Γ -action on V_n).

The cocycle. The two-segment cocycle construction used in the present note originates in personal correspondence from F. Brown to the author [5], which describes the procedure of integrating a weakly holomorphic modular form against the polynomial coefficient system $(X - \tau Y)^n$ along

contours from $\gamma^{-1}\tau_0$ to τ_0 for the generators $\gamma \in \{S, T\}$ of Γ , and extracting periods and quasi-periods from the resulting 1-cocycle modulo coboundaries. Concretely: fix a basepoint $\tau_0 \in \mathbb{H}$ not on the Γ -orbit of any pole of f , and define

$$C_\gamma^f(\tau_0) := (2\pi i)^n \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f = (2\pi i)^n \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}, \quad (12)$$

the contour being any path from $\gamma^{-1}\tau_0$ to τ_0 in $\mathbb{H}$ avoiding the Γ -orbit of the poles of f . The integral is independent of the choice of path within the same homotopy class in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$.

Lemma 2.1 (Cocycle property). *For all $g, h \in \Gamma$,*

$$C_{gh}^f(\tau_0) = C_g^f(\tau_0)|_h + C_h^f(\tau_0). \quad (13)$$

Equivalently, the map $\gamma \mapsto C_\gamma^f(\tau_0)$ is a 1-cocycle of Γ with coefficients in $V_n \otimes \mathbb{C}$.

Proof. Path-additivity gives

$$\int_{(gh)^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} + \int_{h^{-1}\tau_0}^{\tau_0}.$$

In the first integral substitute $\tau = h^{-1}\sigma$, so σ runs from $g^{-1}\tau_0$ to τ_0 . The Γ -equivariance of ω_f converts the integrand to $\omega_f(\sigma)|_h$, yielding the first summand of (13). The second integral gives the second summand directly. $\square$

A cocycle on $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ is determined by its values on the two generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

subject to the relations $S^2 = (ST)^3 = -I$ (which act trivially on V_n for n even). We write

$$C_S^f(\tau_0) = (2\pi i)^n \int_{-1/\tau_0}^{\tau_0} \omega_f, \quad C_T^f(\tau_0) = (2\pi i)^n \int_{\tau_0-1}^{\tau_0} \omega_f, \quad (14)$$

since $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$. These are the two integrals over the contour pair (1).

Moment expansion. Expanding $(X - \tau Y)^n = \sum_{j=0}^n \binom{n}{j} (-\tau)^j X^{n-j} Y^j$ and pulling the polynomial out of the integral defines the moment integrals

$$I_S^j(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^j d\tau, \quad I_T^j(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^j d\tau, \quad (15)$$

for $j = 0, 1, \dots, n$, in terms of which

$$C_S^f(\tau_0) = (2\pi i)^n \sum_{j=0}^n \binom{n}{j} (-1)^j I_S^j(f, \tau_0) X^{n-j} Y^j,$$

and similarly for $C_T^f(\tau_0)$. Theorem 1.1 extracts $\omega^\pm(f)$ from particular linear combinations of these moments.

Basepoint dependence. Varying τ_0 changes C^f by a coboundary, provided f has no interior poles. Define a primitive $F(\tau)$ of ω_f on a simply connected open subset of $\mathbb{H}$ avoiding $\Gamma \cdot f^{-1}(\infty)$. Then for any two basepoints τ_0, τ'_0 in the same connected component, path-additivity gives

$$C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = (P|_\gamma - P) \quad \text{with} \quad P = (2\pi i)^n (F(\tau'_0) - F(\tau_0)) \in V_n \otimes \mathbb{C}. \quad (16)$$

This is the coboundary $\delta P(\gamma) := P|_\gamma - P$. When f has interior poles, dragging τ_0 across a pole picks up a Cauchy residue contribution beyond (16); this wall-crossing structure is the content of Theorem 1.3 in §5.

Cohomology and periods. The cohomology class $[C^f] \in H^1(\Gamma; V_n)$ is independent of τ_0 (on the component) and independent of the choice of δP representative. The classical Eichler–Shimura theorem [12] identifies the cuspidal subspace $H_{\text{cusp}}^1(\Gamma; V_n)$ with $S_k \oplus \overline{S_k}$, and the linear coordinates of $[C^f]$ in this cuspidal subspace are the periods and quasi-periods $\omega^\pm(f), \eta^\pm(f)$ [3].

3 Period extraction: proof of Theorem 1.1

Proof of Theorem 1.1. We work throughout this section at $k = 12$ with $\dim S_k = 1$; the argument at $k \in \{16, 18, 20, 22, 26\}$ is identical with the explicit polynomial coefficients tabulated separately, and the lower even weights $k \in \{2, 4, 6, 8, 10, 14\}$ in the $\dim S_k \leq 1$ family have $S_k = 0$ so the theorem is vacuous. Set $n := k - 2 = 10$.

The coboundary equation on V_n . Pick $f \in S_{12}^!$. Since f has no interior poles, the cocycle $C_\gamma^f(\tau_0)$ of (12) is well-defined for every $\tau_0 \in \mathbb{H}$, and the basepoint-shift coboundary (16) holds without residue corrections (Theorem 1.3 below specialises to local constancy on all of $\mathbb{H}$ in this case). Equivalently, the cohomology class $[C^f] \in H^1(\Gamma; V_n)$ depends on f only.

We seek a polynomial $P = P(\tau_0) \in V_n \otimes \mathbb{C}$ (a coboundary representative depending on τ_0, f) such that the modified T -component $C_T^f(\tau_0) - \delta P(T)$ becomes a fixed polynomial in X, Y with coefficients computable from the Fourier expansion of f . Since $\delta P(T) = P|_T - P$ and T acts by $(X, Y) \mapsto (X + Y, Y)$ on V_n , the operator $\delta_T: P \mapsto P|_T - P$ on V_n shifts the X -variable: writing $P = \sum_{j=0}^n p_j X^{n-j} Y^j$, we have

$$(\delta_T P)(X, Y) = \sum_{j=0}^n p_j [(X + Y)^{n-j} - X^{n-j}] Y^j,$$

which is upper-triangular in the $X^{n-j} Y^j$ basis with zero diagonal entry at $j = n$ (i.e. δ_T annihilates Y^n).

Proposition 3.1 (Bernoulli closed form for δ_T^{-1}). *For every $Q \in V_n$ with Q vanishing at the leading X^n coefficient, there exists $P \in V_n$ unique modulo $\mathbb{C} \cdot Y^n$ with $\delta_T P = Q$. Writing $Q = \sum_{j=1}^n q_j X^{n-j} Y^j$ and $P = \sum_{j=0}^n p_j X^{n-j} Y^j$, the solution is given in closed form by*

$$p_m = \sum_{j=m+1}^n \frac{(n-m)!}{(j-m-1)!(n-j+1)!} \frac{B_{j-m}}{(n-m)!} q_j, \quad (17)$$

for $m = 0, 1, \dots, n-1$, where B_ℓ are Bernoulli numbers; the coefficient p_n is unconstrained and represents the kernel of δ_T .

Proof. The identity $(X + Y)^{n-j} - X^{n-j} = \sum_{\ell=0}^{n-j-1} \binom{n-j}{\ell+1} X^{n-j-\ell-1} Y^{\ell+1}$ gives a strictly upper-triangular matrix for δ_T in the basis $X^{n-m} Y^m$, with zero diagonal at $m = n$. The Bernoulli generating function $\sum_{\ell \geq 0} B_\ell t^\ell / \ell! = t/(e^t - 1)$ is the combinatorial inverse of $\sum_{\ell \geq 1} t^\ell / \ell!$, which is the operator δ_T restricted to monomials. Substituting and collecting coefficients yields (17). $\square$

Period extraction. Apply Proposition 3.1 with $Q = (2\pi i)^{-n} C_T^f(\tau_0)$. Since $f \in S_{12}^!$ has vanishing constant term $c_f(0) = 0$, the X^n coefficient of C_T^f vanishes, so the hypothesis of Proposition 3.1 is satisfied. Let $P(\tau_0) \in V_n$ be the resulting solution (modulo $\mathbb{C}Y^n$); the coboundary δP encodes the basepoint dependence of C^f . By (16) the modified cocycle $C^f - \delta P \cdot (2\pi i)^n$ is basepoint-independent, and its evaluation on S gives the Eichler–Shimura period polynomial

$$r_f(X, Y) := (C_S^f(\tau_0) - (2\pi i)^n \delta P(\tau_0)(S)) \in V_n, \quad (18)$$

where $\delta P(\tau_0)(S) = P(\tau_0)|_S - P(\tau_0)$ and $P|_S$ has $(X, Y) \mapsto (-Y, X)$. Brown’s normalisation [3] identifies the linear coordinates of r_f with the periods and quasi-periods $\omega^\pm(f), \eta^\pm(f)$: at $k = 12$,

$$r_f(X, Y) = \omega^+(f) p_f^+(X, Y) + \omega^-(f) p_f^-(X, Y) + (\text{Eisenstein-direction terms in } \eta^\pm),$$

where $p_f^\pm \in V_n$ are the standard cuspidal even/odd projection polynomials of [12, § ...].

The kernels $k_S(j, \tau)$ and $k_T(j, \tau)$. Equation (18) expressed in terms of the moment integrals (15) reads

$$r_f(X, Y) = (2\pi i)^n \sum_{j=0}^n \binom{n}{j} (-1)^j \left[I_S^j(f, \tau_0) - I_T^{\delta P, j}(f, \tau_0) \right] X^{n-j} Y^j,$$

where $I_T^{\delta P, j}$ packages the $\delta P(S)$ contribution into a T -segment integral against the Bernoulli-polynomial kernel from (17). Reading off the coefficient of $X^{n-j} Y^j$ for each $j \in \{2, \dots, n\} = \{2, \dots, k-2\}$ (the indices corresponding to monomials lying in the cuspidal period polynomial space; $j = 0, 1$ correspond to the Eisenstein direction and the δ_T kernel respectively) gives (5) with

$$k_S(j, \tau) = \tau^{j-1}, \quad (19)$$

$$k_T(j, \tau) = \frac{1}{\binom{n}{j-1}} \left[\sum_{m=0}^{n-1} \frac{(n-m)!}{(j-m-1)! (n-j+1)!} \frac{B_{j-m}}{(n-m)!} \tau^m \right], \quad (20)$$

the Bernoulli-polynomial-built T -kernel of degree at most $n-1$ in τ . Both kernels are $\mathbb{Q}$ -polynomial. The S -kernel $k_S(j, \tau) = \tau^{j-1}$ is the integer- $s = j$ specialisation of the Mellin kernel τ^{s-1} in (6). The T -kernel $k_T(j, \tau)$ matches the integer- $s = j$ specialisation of the Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3), via the collapse $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ ($m \geq 0$); this identification is the bridge between Theorem 1.1 and Theorem 1.2, and is made explicit in the worked $k = 12$ subsection of §4.

Lockstep and basepoint independence. Both follow from (18): the right side is a fixed polynomial in V_n , independent of both τ_0 (by the coboundary cancellation) and j (since reading off different coefficients of the same polynomial returns different multiples of the same $\omega^\pm$). The factor $\binom{k-2}{j}^{-1} (2\pi i)^{k-1}$ in (5) normalises so that the right side equals $\omega^\pm(f)$ for each cuspidal $j \in \{2, \dots, k-2\}$, independently of j . At $k = 12$ this yields a 9-fold redundant extraction of $\omega^\pm(f)$ from the Fourier coefficients of f , with the redundancy reflecting the 1-dimensionality of the cuspidal period-polynomial direction in V_{10} for the running case $f \in \{\Delta, \widehat{\Delta}\}$. $\square$

4 The Hurwitz-zeta extension and the L -function on $M_k^!$: proof of Theorem 1.2

4.1 Derivation of the kernel $\tilde{k}_T$

We seek a kernel $\tilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that $L^*(f, s; \tau_0)$ as defined in (6) is independent of τ_0 for every $f \in M_k^!$. Differentiating (6) in τ_0 and using $\Delta(-1/\tau_0) = \tau_0^k \Delta(\tau_0)$ (the S -modularity of f at the S -segment endpoint $-1/\tau_0$) together with $f(\tau_0 - 1) = f(\tau_0)$ (the T -modularity at $\tau_0 - 1$) gives, after cancelling the common factor $i^{-s} f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s}. \quad (21)$$

The right side picks up the S -segment endpoint contribution at $-1/\tau_0 = e^{i\pi} \tau_0^{-1}/(-1) = e^{i\pi}/\tau_0$ via the branch identity $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ combined with the Jacobian $d(-1/\tau_0)/d\tau_0 = \tau_0^{-2}$ and the modularity factor τ_0^k , giving $e^{i\pi(s-1)} \tau_0^{k-1-s}$ after simplification.

Equation (21) is the complex- s analog of the polynomial coboundary equation $\delta_T P = Q$ of §3. The Hurwitz raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma}$$

inverts the shift operator $\tau \mapsto \tau-1$ for any complex σ , on the analytic function $\zeta(\sigma, \tau+1)$. Applying at $(\sigma, a) = (1-s, \tau_0)$ and $(\sigma, a) = (s-(k-1), \tau_0)$, and combining with the sign $e^{i\pi(s-1)}$ gives (3).

Lemma 4.1. *For $f \in M_k^!$, $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ defined by (6) is independent of $\tau_0 \in \mathbb{H}$ for every $s \in \mathbb{C}$.*

Proof. Direct τ_0 -differentiation, using (21) to cancel the boundary contributions from the two segments. Both integrals in (6) are finite at every $\tau_0 \in \mathbb{H}$ because f has no poles in $\mathbb{H}$ and the kernels τ^{s-1} , $\tilde{k}_T(\tau, s)$ are analytic in $\tau \in \mathbb{H}$ for every fixed $s \in \mathbb{C}$ (the Hurwitz $\zeta(\sigma, a)$ is analytic in a on $\text{Re}(a) > 0$). $\square$

4.2 The Hurwitz-unfolding lemma

Lemma 4.2 (Hurwitz unfolding). *For every positive integer n and every $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (22)$$

Proof. First assume $\text{Re}(\nu) > 1$, where the Hurwitz series $\zeta(\nu, \sigma) = \sum_{m \geq 0} (\sigma + m)^{-\nu}$ converges uniformly on the compact segment $[i, i+1]$ [6, §25.11]. Series-integral exchange and the substitution $u = \sigma + m$ in the m -th summand collapse the sum to a single integral by integer- n periodicity $e^{2\pi i n m} = 1$:

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = \sum_{m \geq 0} \int_{i+m}^{i+m+1} e^{2\pi i n u} u^{-\nu} du = \int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du,$$

an integral along the horizontal ray $\{i+t : t \geq 0\} \subset \mathbb{H}$.

Substitute $t = -2\pi i n u$: $u = it/(2\pi n)$, $du = i dt/(2\pi n)$, and $u^{-\nu} = i^{-\nu} (2\pi n)^{\nu} t^{-\nu}$ on the principal branch (consistent with $\arg(u) = \pi/2$ at $u = i$ and $\arg(t) = 0$ at $t = 2\pi n$). The contour in t runs from $2\pi n$ down to $2\pi n - i\infty$, in the right half-plane. Since $e^{-t} t^{-\nu}$ decays exponentially as $|t| \rightarrow \infty$ in $\text{Re}(t) > 0$, Cauchy's theorem applied to the rectangle $\{2\pi n \leq \text{Re}(t) \leq R, -R \leq$

$\text{Im}(t) \leq 0\}$ in the limit $R \rightarrow \infty$ deforms the vertical contour $[2\pi n, 2\pi n - i\infty]$ to the real-axis contour $[2\pi n, +\infty]$:

$$\int_{2\pi n}^{2\pi n - i\infty} e^{-t} t^{-\nu} dt = \int_{2\pi n}^{\infty} e^{-t} t^{-\nu} dt = \Gamma(1 - \nu, 2\pi n)$$

by the integral definition of the upper incomplete gamma [6, §8.2]. Collecting prefactors gives (22) for $\text{Re}(\nu) > 1$.

Both sides of (22) are entire functions of ν : the upper incomplete gamma $\Gamma(s, x)$ is entire in s for fixed $x \neq 0$ [6, §8.2.7], and the integral on the left is entire in ν because $\zeta(\sigma, \sigma_0)$ is meromorphic in σ with a simple pole only at $\sigma = 1$, whose residue contribution to $\int_{i-1}^i e^{2\pi i n \tau} d\tau = 0$ for nonzero integer n (since $e^{2\pi i n(i-1)} = e^{2\pi i n \cdot i}$). Analytic continuation in ν extends (22) to all complex ν . $\square$

4.3 Per-mode evaluation: proof of Theorem 1.2

By Lemma 4.1, the value $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) = L^*(f, s)$ is common to all admissible contours and basepoints. At the elliptic fixed-point basepoint $\tau_0 = i$, the S -segment $[-1/i, i] = [i, i]$ has zero length and the S -integral in (6) vanishes identically, leaving

$$L^*(f, s; \gamma_S(i) = 0, \gamma_T(i)) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau.$$

Expanding $f = \sum_n c_f(n) q^n$ as a q -series and swapping sum and integral (justified by uniform convergence of the q -series on the compact segment $[i-1, i]$, including any finite number of polar modes since $|q^{-1}| = e^{2\pi}$ is bounded there) gives

$$L^*(f, s) = i^{-s} \sum_n c_f(n) \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau. \quad (23)$$

We treat the constant Fourier mode $n = 0$ and the non-constant modes $n \neq 0$ separately.

The constant Fourier mode. For $n = 0$, the integrand reduces to $\tilde{k}_T(\tau, s)$ itself. The Hurwitz zeta has antiderivative $\int \zeta(\sigma, a) da = -\zeta(\sigma - 1, a)/(\sigma - 1) + C$ for $\sigma \neq 1$, and the Hurwitz raising identity $\zeta(\sigma - 1, a + 1) - \zeta(\sigma - 1, a) = -a^{1-\sigma}$ at $a = i$ gives

$$\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma - 1}. \quad (24)$$

Applying (24) to each half of $\tilde{k}_T$ at $\sigma = 1 - s$ and $\sigma = s - (k - 1)$:

$$\int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -\frac{i^s}{s} + e^{i\pi(s-1)} \frac{i^{k-s}}{k-s}.$$

The overall i^{-s} prefactor and the phase collapse $i^{-s} e^{i\pi(s-1)} i^{k-s} = -i^k$ give the constant-term contribution

$$c_f(0) \cdot i^{-s} \int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right). \quad (25)$$

This matches the polar term of [2, Lemma 2.1] (equivalently, the polar piece of [1, Thm. 1.1]) at the splitting point $B = 1$.

The non-constant Fourier modes. For $n \neq 0$, substituting $\sigma = \tau + 1$ shifts the contour to $[i, i + 1]$ with the prefactor $e^{-2\pi i n} = 1$. Apply Lemma 4.2 to each half of $\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)}\zeta(s - (k - 1), \tau + 1)$, i.e. at $\nu = 1 - s$ and $\nu = s - (k - 1)$:

$$\begin{aligned} \int_i^{i+1} e^{2\pi i n \sigma} \zeta(1 - s, \sigma) d\sigma &= i^s (2\pi n)^{-s} \Gamma(s, 2\pi n), \\ \int_i^{i+1} e^{2\pi i n \sigma} \zeta(s - (k - 1), \sigma) d\sigma &= i^{k-s} (2\pi n)^{s-k} \Gamma(k - s, 2\pi n). \end{aligned}$$

The first half against the i^{-s} prefactor gives $\Gamma(s, 2\pi n)/(2\pi n)^s$; the second half against the relative phase $-e^{i\pi(s-1)}$ collapses via $-i^{-s} e^{i\pi(s-1)} i^{k-s} = i^k$ to $i^k \Gamma(k - s, 2\pi n)/(2\pi n)^{k-s}$. Adding the two halves gives the per-mode summand

$$c_f(n) \cdot i^{-s} \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau = c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k - s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (26)$$

which is the $B = 1$ value of the cuspidal summand of [2, Lemma 2.1].

Summing (25) and (26) over all Fourier modes reproduces the closed-form sum (7).

4.4 Worked example at $k = 12$: Bernoulli polynomials and numerics

At $k = 12$ the kernel of (3) is

$$\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - 11, \tau + 1).$$

At integer values of $s \in \{1, 2, \dots, 11\}$, both Hurwitz zeta factors collapse to Bernoulli polynomials via $\zeta(-m, a) = -B_{m+1}(a)/(m + 1)$ ($m \geq 0$), and $\tilde{k}_T(\tau, s)$ becomes a polynomial of degree ≤ 11 in τ . Three representative values:

$$\begin{aligned} \tilde{k}_T(\tau, 1) &= -\tau - \frac{1}{2} + \frac{1}{11} B_{11}(\tau + 1), & \tilde{k}_T(\tau, 6) &= -\frac{1}{3} B_6(\tau + 1), \\ \tilde{k}_T(\tau, 11) &= -\frac{1}{11} B_{11}(\tau + 1) + \tau + \frac{1}{2}. \end{aligned}$$

The $s \leftrightarrow 12 - s$ symmetry follows from the $s \leftrightarrow k - s$ structure of the two-term kernel (with $i^k = +1$ at $k = 12$).

Identification with the polynomial kernels of Theorem 1.1. The T -kernels $k_T(j, \tau)$ of (19) and (??) are precisely the integer- $s = j$ specialisations of $\tilde{k}_T(\tau, s)$, up to the normalisation $\binom{k-2}{j-1}^{-1}$ that converts the L -value convention (governing $\tilde{k}_T$) to the $\omega^\pm$ convention (governing $k_T(j, \tau)$). Explicitly, for $j \in \{2, \dots, 10\}$ at $k = 12$,

$$k_T(j, \tau) = \frac{(-1)^{j-1}}{\binom{10}{j-1}} \tilde{k}_T(\tau, j),$$

exhibiting the polynomial period-kernel construction of Theorem 1.1 as the integer- s shadow of the Hurwitz-zeta kernel of Theorem 1.2.

Numerical values. Running the companion `mpmath` scripts [4] at 50-digit precision, with $\tau_0 \in \{0.3 + 1.2i, i, 1.5i, -0.4 + 0.8i\}$ and $f \in \{\Delta, \hat{\Delta}\}$, reproduces the values of $\omega^\pm(\Delta)$ and $\eta^\pm(\hat{\Delta})$ of [3]:

$$\begin{aligned} \omega^+(\Delta) &\approx -68916772.8096\dots, & \omega^-(\Delta)/i &\approx -5585015.3793\dots, \\ \eta^+(\hat{\Delta}) &\approx 1.27202 \times 10^{11}, & \eta^-(\hat{\Delta})/i &\approx 1.02767 \times 10^{10}, \end{aligned}$$

with relative residuals $\lesssim 10^{-40}$, exhibiting lockstep across $j \in \{2, \dots, 10\}$ and basepoint-independence across all four τ_0 choices.

5 Wall-crossing for meromorphic forms: proof of Theorem 1.3

5.1 Existence and local constancy

Let f be a meromorphic modular form of even weight k on Γ , and let $\tau_0 \in \mathbb{H}$ be a basepoint such that the contour pair (1) avoids the Γ -orbit of $f^{-1}(\infty) \cap \mathbb{H}$. The integrands $f(\tau) \tau^{s-1}$ and $f(\tau) \tilde{k}_T(\tau, s)$ are continuous on each compact segment for every fixed $s \in \mathbb{C}$ (the kernels are analytic in $\tau \in \mathbb{H}$), so both integrals in (2) are finite and $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ is well-defined. This proves Theorem 1.3(a).

For local constancy, the argument of §4.1 applies verbatim: differentiating (2) in τ_0 gives the boundary contributions

$$i^{-s} f(\tau_0) \left[\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) + \tau_0^{s-1} - e^{i\pi(s-1)} \tau_0^{k-1-s} \right] = 0$$

by (21), provided the modularity identities $f(-1/\tau_0) = \tau_0^k f(\tau_0)$ and $f(\tau_0 - 1) = f(\tau_0)$ hold at the contour endpoints (which they do for every meromorphic modular form, irrespective of the location of interior poles). Hence $dL^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))/d\tau_0 = 0$ whenever the contour pair varies within the admissible open subset of $\mathbb{H}$. This establishes Theorem 1.3(b).

5.2 Wall-crossing across an interior pole

Suppose a continuous one-parameter deformation of τ_0 to τ'_0 in $\mathbb{H}$ moves the S -segment $[-1/\tau_0, \tau_0]$ across an interior pole $\tau_p \in \mathbb{H}$ of f (the T -segment case is identical up to the choice of kernel; see below). Concretely, let $\Sigma_t \subset \mathbb{H}$ denote the S -segment at parameter $t \in [0, 1]$, with Σ_0 the S -segment at τ_0 and Σ_1 at τ'_0 , and assume that the swept-out region $\bigcup_t \Sigma_t$ encloses τ_p with winding number $\sigma_p^S \in \{+1, -1\}$ (orientation of the crossing). By Cauchy's theorem applied to the closed contour $\Sigma_1 - \Sigma_0$,

$$\int_{\Sigma_1} f(\tau) \tau^{s-1} d\tau - \int_{\Sigma_0} f(\tau) \tau^{s-1} d\tau = 2\pi i \sigma_p^S \operatorname{Res}(f, \tau_p) \tau_p^{s-1}$$

for a simple pole of f at τ_p . An identical argument applies when the T -segment $[\tau_0 - 1, \tau_0]$ sweeps across τ_p , yielding

$$\int_{T\text{-seg at } \tau'_0} - \int_{T\text{-seg at } \tau_0} = 2\pi i \sigma_p^T \operatorname{Res}(f, \tau_p) \tilde{k}_T(\tau_p, s),$$

where $\sigma_p^T \in \{+1, -1\}$ encodes the orientation of the T -segment crossing. Combining (with the i^{-s} prefactor) gives

$$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) - L^*(f, s; \gamma_S(\tau'_0), \gamma_T(\tau'_0)) = -2\pi i i^{-s} \operatorname{Res}(f, \tau_p) \left[\sigma_p^S \tau_p^{s-1} + \sigma_p^T \tilde{k}_T(\tau_p, s) \right]. \quad (27)$$

This is the residue formula $\mathcal{R}_{\tau_p}(f, s)$ of (10), with the winding-number indices σ_p^S, σ_p^T making explicit the dependence on which segment(s) cross. At a generic wall only one of σ_p^S, σ_p^T is nonzero; both can be nonzero on the codimension-two loci where the S - and T -segments simultaneously cross Γ -images of τ_p .

For a pole of order $m \geq 1$ at τ_p with Laurent expansion $f(\tau) = \sum_{\ell \geq -m} r_{f, \tau_p}(\ell) (\tau - \tau_p)^\ell$, the residue formula (27) generalises by replacing $\operatorname{Res}(f, \tau_p) K(\tau_p, s)$ with $\sum_{\ell=1}^m r_{f, \tau_p}(-\ell) \partial_{\tau_p}^{\ell-1} K(\tau_p, s) / (\ell-1)!$ for each relevant kernel $K \in \{\tau^{s-1}, \tilde{k}_T(\tau, s)\}$, by the standard residue formula for higher-order poles.

5.3 Reduction on regular-at-cusps meromorphic forms

Let $f \in F_k$ be regular at the cusp $i\infty$, so $f(\tau) = \sum_{n \geq N} c_f(n) q^n$ for some $N \geq 0$.

Lemma 5.1 (Cusp decay of the T -segment). *For $f \in F_k$ regular at the cusp with $f(\tau) = O(q^N)$ at $i\infty$ for $N \geq 1$, every $s \in \mathbb{C}$, and every $\varepsilon \in \mathbb{R}$ fixed,*

$$\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = O(|\tau_0|^{C(\operatorname{Re} s)} e^{-2\pi N \operatorname{Im}(\tau_0)}) \quad \text{as } \operatorname{Im}(\tau_0) \rightarrow \infty, \operatorname{Re}(\tau_0) = \varepsilon.$$

The exponent $C(\operatorname{Re} s) = \max(0, 1/2 - \operatorname{Re}(1-s)) + \max(0, 1/2 - \operatorname{Re}(s - (k-1))) + \varepsilon_0$ for any $\varepsilon_0 > 0$ is controlled by the standard polynomial-growth bound for the Hurwitz zeta on vertical strips [6, §25.11.36]. When $N = 0$ (i.e. f has a nonzero constant term at $i\infty$), the T -segment is $O(|\tau_0|^{C(\operatorname{Re} s)})$ but does not decay; in this case the constant-term Fourier mode must be split off and treated separately (see [2, Lemma 2.1] for the analogous polynomial correction in the deformed-Mellin construction).

Proof. On the segment $\tau \in [\tau_0 - 1, \tau_0]$ with $\operatorname{Im}(\tau) = \operatorname{Im}(\tau_0)$, the bound $|f(\tau)| \leq M e^{-2\pi N \operatorname{Im}(\tau_0)}$ holds for some $M = M(f)$ depending only on f . Combining with the polynomial-in- τ growth of $\tilde{k}_T(\tau, s)$ from [6, §25.11.36] and integrating over the length-1 segment gives the stated bound. $\square$

To match the deformed-Mellin construction of $L_f^*(s)_{\text{int}}$ (see [2]), take $\tau_0 = \varepsilon + i\Lambda$ for some small $\varepsilon \in \mathbb{R}$ chosen so that $\operatorname{Re}(\tau_0) = \varepsilon$ avoids all Γ -images of interior poles of f . As $\Lambda \rightarrow \infty$:

- The T -segment integral $\rightarrow 0$ by Lemma 5.1 (using $N \geq 1$; for $N = 0$ the analog argument with the constant-term subtraction $c_f(0)$ from [2] applies).
- The S -segment $[-1/\tau_0, \tau_0]$ approaches the $\Lambda \rightarrow \infty$ limit of the Λ -truncated Mellin contour along $\operatorname{Re}(\tau) = \varepsilon$ (modulo orientation, which fixes the L_{int}^* convention).

By Lemma 4.1 (extended to meromorphic f within a fixed connected component, by §5.1 above), the left side is Λ -independent, so

$$L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) = \lim_{\Lambda \rightarrow \infty} L^*(f, s; \gamma_S(\varepsilon + i\Lambda), \gamma_T(\varepsilon + i\Lambda)) = L_f^*(s)_{\text{int}, \gamma_\varepsilon},$$

where γ_ε is the deformed-Mellin contour along $\operatorname{Re}(\tau) = \varepsilon$. This proves Corollary 1.4. $\square$

5.4 Remark on the polylog-projection machinery

The polylog projection $\hat{P}_{\tau_p} : f \mapsto \sum_{m > 0} r_{f, \tau_p}^*(m) \operatorname{Li}_{1-m}(e(\tau - \tau_p))$ (see [2, §2.2]) subtracts the leading polar tails of f at each interior pole τ_p , ensuring that the Λ -truncated Mellin integral against f has a finite $\Lambda \rightarrow \infty$ limit (necessary in the deformed-Mellin construction because of the Hagedorn-rate Fourier coefficients $|c_f(n)| \sim n^{p-1} e^{2\pi n \operatorname{Im}(\tau_p)}$ generated by interior poles [2, Prop. 1.1, 2.7]). In the finite-contour formulation (2) the $\Lambda \rightarrow \infty$ limit is not taken, so $\hat{P}_{\tau_p}$ is not required for the existence or local-constancy properties of L^* . The polylog-projection machinery does remain relevant, however, for the *closed-form per-mode evaluation* of L^* in incomplete gammas parallel to (7), where the Hagedorn-rate Fourier modes must be split off and dealt with separately before each mode contributes a clean incomplete-gamma term. The systematic interior-pole closed form parallel to Theorem 1.2 is left to the sequel.

6 Outlook

Wilson-loop / holonomy interpretation. The period-defining contour pair, projected to the modular curve $Y(1)$, descends to the two generators of the orbifold fundamental group $\pi_1^{\text{orb}}(Y(1)) \cong \mathbb{Z}/2 * \mathbb{Z}/3$. The polynomial coefficient system V_{k-2} is a flat local system over $Y(1)$, and the cocycle C_γ^f is the weighted holonomy of this flat connection along the loop γ , with f supplying the gauge-invariant weight. The cohomology class $[C^f] \in H^1(\Gamma; V_{k-2})$ is the analog of a gauge-invariant observable; τ_0 is the analog of a trivialisation frame; $\omega^\pm(f)$ are coordinates of $[C^f]$ in the cuspidal subspace, the analog of physical observables read off a Wilson-loop expectation. The period-defining finite-contour functional is, in this dictionary, the modular-form analog of a Wilson-loop holonomy on the modular curve.

The on-shell methods analogy. The classical Mellin construction is “off-shell” in the same sense as a Lagrangian gauge-theory amplitude calculation: the intermediate integrand $f(\tau) \tau^{s-1} d\tau$ encodes gauge-redundant information (the basepoint τ_0 , the contour to the cusp), and the gauge-invariant observable $L^*(f, s)$ is reached only after a non-canonical limit. Theorem 1.2 is an “on-shell” reformulation in which the gauge-invariant observable is the value of a finite-contour functional at any admissible basepoint, with no limit and no regulator. The modular-form analog of the on-shell S-matrix programme [14, 15] — constructing $\omega^\pm$ and $L^*(f, s)$ from $\{c_f(n)\}$ via a kernel that does not pass through the cocycle / coboundary scaffolding — is partially answered by Theorem 1.1 on the period side and Theorem 1.2 on the L -function side, with the remaining question being whether the kernels themselves admit a construction independent of the δ_T^{-1} Bernoulli operator.

Regularisation as a symptom of bad framing. The disappearance of any regularisation step in (2) is an instance of a recurring methodological pattern: *when a divergence arises only after one has made a non-canonical upstream choice, finding the canonical formulation eliminates the divergence together with the regulator.* The same move underlies the on-shell amplitudes programme [14]. The pattern is not universal — chiral anomalies in even dimensions are a counterexample, with cohomologically robust “regulator” content invariant under reframings — but it is a useful prior, and the finite-contour formulation here is a particularly clean instance in which the canonical $L^*(f, s)$ falls out as the value of a well-defined linear functional at any admissible basepoint, with no auxiliary parameter to take to infinity and no contour to deform to a cusp.

The geometric content of the Mellin kernel. The polynomial coefficient system V_{k-2} underlying the period construction is the symmetric-power local system that gives the polynomial kernel $(X - \tau Y)^{k-2}$ of the period polynomial. Its flat connection on the modular curve is the natural geometric home of $\omega^\pm(f)$. The Hurwitz-zeta kernel $\tilde{k}_T(\tau, s)$ of (3) extends the polynomial kernel to a one-parameter analytic family in s , but the geometric content of the bare Mellin kernel $f(\tau) \tau^{s-1} d\tau$ itself — as opposed to its specialisation at integer s to a polynomial-in- V_{k-2} kernel — is not visible in the construction of this paper. A natural conjecture is that $\tau^{s-1} d\tau$ is a section of an infinite-dimensional principal-series-type local system on $Y(1)$, in the same one-parameter family as the Lewis–Zagier period functions for Maass wave forms [13]. Identifying the relevant local system explicitly, and connecting the Hurwitz-zeta inversion of δ_T to a coboundary statement in this larger system, is an open question.

Towards systematic interior-pole L -function theory. Theorem 1.3 establishes existence and wall-crossing for $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0))$ on arbitrary meromorphic f , but does not give a closed-form per-mode evaluation parallel to the incomplete-gamma sum (7). Developing the systematic per-mode theory — in particular adapting the polylog-projection machinery $\hat{P}_{\tau_p}$ (see [2]) to the finite-contour setting and matching the wall-crossing data (9) to the iterated-Eichler-integral framework of Brown–Hain on $\mathcal{M}_{1,1}$ and the universal elliptic curve [3] — is the natural sequel. The arithmetic of the wall-crossing residues $\mathcal{R}_{\tau_p}(f, s)$ for forms with poles at CM points (e.g. $\tau_p \in \{i, \rho\}$) is a particularly clean target, where the Laurent coefficients of f take values in number fields and $\mathcal{R}_{\tau_p}$ acquires explicit arithmetic structure.

A Algebraic derivation of the Hurwitz-zeta kernel

The body of §4.1 derives $\tilde{k}_T(\tau, s)$ by differentiating the two-segment functional in τ_0 and demanding the result vanish. Here we record the parallel derivation that proceeds *from* the classical Mellin transform of a holomorphic cusp form, splitting it at the contour endpoints $\{-1/\tau_0, \tau_0\}$ and demanding the resulting tail be representable as a T -segment integral. Both derivations produce the same shift-difference equation and the same kernel; this appendix makes the algebraic provenance of the Hurwitz-zeta combination manifest.

Step 1: Mellin split at the contour endpoints. Let $f \in S_k$ with k even. The classical completed L -function $L^*(f, s) = (2\pi)^{-s} \Gamma(s) L(f, s)$ has integral representation

$$L^*(f, s) = -i \int_0^{i\infty} f(\tau) (\tau/i)^{s-1} d\tau = i^{-s} \int_0^{i\infty} f(\tau) \tau^{s-1} d\tau,$$

the second equality using $(\tau/i)^{s-1} = i^{1-s} \tau^{s-1}$ on the imaginary axis. Pick $\tau_0 \in \mathbb{H}$ and split the imaginary-axis contour at the two endpoints $\{-1/\tau_0, \tau_0\}$ in (1). Reorganising the three resulting pieces using S -modularity $f(-1/\tau) = \tau^k f(\tau)$ on the lower segment $[0, -1/\tau_0]$ (via the substitution $\tau \mapsto -1/\tau$) and computing the q -series-against- τ^{s-1} integral on $[\tau_0, i\infty]$ directly yields, for every $\tau_0 \in \mathbb{H}$ and every $s \in \mathbb{C}$,

$$L^*(f, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (28)$$

The left side is τ_0 -independent (it is the classical L -value); both terms on the right side individually depend on τ_0 , but their sum does not. At $\tau_0 = i$ the argument $-2\pi i n \tau_0 = 2\pi n$ is real and (28) reduces to the standard BFK incomplete-gamma sum of [1].

The right side of (28) contains the S -segment integral $i^{-s} \int_{-1/\tau_0}^{\tau_0}$ explicitly; the remaining incomplete-gamma sum plays the role that we want to be a T -segment integral.

Step 2: ansatz for the T -segment kernel. Postulate that there exists a kernel $\tilde{k}_T(\tau, s)$, analytic in $\tau \in \mathbb{H}$ and meromorphic in $s \in \mathbb{C}$, such that

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = \sum_{n \geq 1} c_f(n) \left[\frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right] \quad (29)$$

for every $f \in S_k$ and every $\tau_0 \in \mathbb{H}$. If such a kernel exists, substituting (29) into (28) gives the two-segment functional $L^*(f, s; \gamma_S(\tau_0), \gamma_T(\tau_0)) = i^{-s} [\int_{\gamma_S} + \int_{\gamma_T}]$ as a τ_0 -independent expression equal to $L^*(f, s)$.

Step 3: τ_0 -derivative gives a shift-difference equation. Differentiate both sides of (29) in τ_0 . On the left, the only τ_0 -dependence is in the integration limits, so $d/d\tau_0$ acts as

$$\frac{d}{d\tau_0} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = f(\tau_0) [\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s)],$$

using $f(\tau_0 - 1) = f(\tau_0)$ for the lower limit. On the right of (29), differentiating each incomplete gamma in its second argument (the explicit τ_0 -dependence) and assembling the Fourier series yields, after canceling a common factor of $f(\tau_0)$,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s} \quad \text{for all } \tau_0 \in \mathbb{H}. \quad (30)$$

The first term on the right is the contribution of the τ^{s-1} -Mellin endpoint at τ_0 ; the second term is the S -modularity image of the same endpoint at $-1/\tau_0$, carrying the branch factor $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$ together with the modularity Jacobian τ_0^k . Equation (30) is a shift-difference equation of order one in τ , with a sum of two power-of- τ terms on the right side.

Step 4: Hurwitz raising solves the shift-difference equation. The Hurwitz zeta function $\zeta(\sigma, a)$ satisfies the raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma} \quad (\sigma \in \mathbb{C}, \operatorname{Re}(a) > 0) \quad (31)$$

(see [6, §25.11.3]). Applied at $(\sigma, a) = (1-s, \tau)$ and $(\sigma, a) = (s-(k-1), \tau)$ respectively, this gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-(k-1), \tau+1) - \zeta(s-(k-1), \tau) = -\tau^{k-1-s}.$$

Combining with the relative phase $e^{i\pi(s-1)}$ from the second term of (30) gives the kernel

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1), \quad (32)$$

which by construction satisfies (30) and hence (29). The kernel is analytic in τ on $\mathbb{H}$ (both Hurwitz zetas have second argument $\tau+1$ with $\operatorname{Re}(\tau+1) > 0$) and meromorphic in s with simple poles only at $s=0$ and $s=k$, where the per-mode integrals collect removable cancellations (cf. Lemma 4.2). Equation (32) is the kernel (3) of the main text.

Where DR2022 stops. The kernel (32) has two Hurwitz-zeta terms. The construction of [8, Thm. 1.2] gives the single-term

$$\tilde{k}_T^{\text{DR}}(\tau, m) = \zeta^*(1-m, \tau+1)$$

on the segment $[i, i+1]$, at integer m , which is the first term of (32). The structural origin of this gap is the absence of an S -segment in their construction: without the S -modularity-transformed contribution at the $-1/\tau_0$ endpoint in Step 3, the shift-difference equation (30) has only the single right-hand-side term $-\tau_0^{s-1}$, and its solution is the single-Hurwitz kernel. The second $e^{i\pi(s-1)} \tau_0^{k-1-s}$ term appears precisely because the two-segment functional includes the S -segment and its endpoint at $-1/\tau_0$ contributes under τ_0 -differentiation via S -modularity.

B Eichler–Shimura for physicists

This appendix is intended as an exposition aid for readers approaching the subject from theoretical physics. It is removable without affecting the logical content of the paper and is excluded from the journal/arXiv version.

[To be filled in. Topics: cusp forms vs. holomorphic vs. weakly holomorphic vs. meromorphic modular forms $(S_k, M_k, M_k^!, F_k)$; the upper half-plane $\mathbb{H}$ and the modular curve $Y(1) = \Gamma \backslash \mathbb{H}$ as the moduli space $\mathcal{M}_{1,1}$ of elliptic curves; the polynomial coefficient system V_{k-2} and the flat connection on the modular curve; the cocycle / coboundary language of $H^1(\Gamma; V_{k-2})$ and its cuspidal subspace H_{cusp}^1 ; periods and quasi-periods $\omega^\pm, \eta^\pm$ via Eichler integrals; physical analogies (worldsheet, polarization tensors on auxiliary variables, holonomies of flat connections, gauge-invariant observables). Worked computations at $k = 12$ where helpful. $\approx 2\text{--}3$ pages, terse.]

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- [4] Companion verification scripts. `verify_s6.py`, `verify_multi_s.py`: per-mode verification of Theorem 1.2 at 50-digit `mpmath` precision. Additional scripts for Theorem 1.3 and Theorem 1.1 numerics to be added.
- [5] F. Brown, personal communication to the author, 28 June 2019. This correspondence describes the two-segment cocycle procedure (integrate along contours from $\gamma^{-1}\tau_0$ to τ_0 for the two generators $\gamma \in \{S, T\}$ of Γ , modify by a coboundary $P|(1 - T)$ to kill the T -component, then read $\omega^\pm$ and $\eta^\pm$ off the S -component), and is the structural starting point of the construction in the present note. An accompanying unpublished note on quasi-periods of cusp forms of levels 6 and 8 was attached to the correspondence.
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